higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 3

(12041003)

26 February 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages and a formula sheet of 2 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

1.1 An atom is the smallest unit of matter that retains all of the chemical properties of an element.

1.1.1 Briefly explain the structure of an atom. (3)

1.1.2 Explain the term *atomic mass*. (1)

1.2 Different materials react differently when temperature is applied.

Explain the effect of temperature on the resistance of a germanium semiconductor. (2)

1.3 Explain the difference between *current* and *electrons* as used in the electrical field. (2)

1.4 A coil has a resistance of 10 ohms at 20 °C.

What will the resistance be at a temperature of 60 °C if the temperature coefficient of resistance at 20 °C is 0,0039 per °C? (3)

1.5 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.5.1–1.5.4) in the ANSWER BOOK.

1.5.1 When the cross-sectional area of a conductor is doubled, its resistance will be halved.

1.5.2 Free electrons are electrons in the outer shell of an atom.

1.5.3 The resistance of alloys increases as their temperature increases.

1.5.4 In an atom the number of protons is less than the number of electrons.

(4 × 1) (4)

[15]

QUESTION 2

2.1 The magnetic field in an electromagnet is produced by an electric current.

List THREE applications of electromagnets. (3)

2.2 Explain the term *magnetic field strength*. (2)

2.3 Calculate the flux density when the flux is 800 µWb and the area is 0,004 m^2. (2)

2.4 Calculate the magnitude of force exerted on a conductor that is 15 cm long and carries a current of 4 A when it lies at:

2.4.1 90° to the magnetic field having a flux density of 450 mT

2.4.2 60° to the magnetic field having a flux density of 300 mT

(2 × 2) (4)

2.5 Use the table below to compare electrical and magnetic quantities by writing only the quantity next to the question number (2.5.1–2.5.4) in the ANSWER BOOK.

ELECTRIC CIRCUITS	MAGNETIC CIRCUITS
e.g. emf E (V)	mmf (A)
2.5.1	2.5.2
2.5.3	2.5.4

(4 × 1) (4)

2.6 Hans Christian Oersted discovered the effect of an electric current on a magnetic field.

Explain how magnetic fields are produced by electric currents. (3)

2.7 When an emf is generated in a coil by a change in magnetic flux, its polarity is such that it produces a current of which the magnetic field opposes the change of magnetic flux responsible for inducing it.

Name TWO methods used to determine the direction of the induced emf. (2)

[20]

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QUESTION 3

3.1

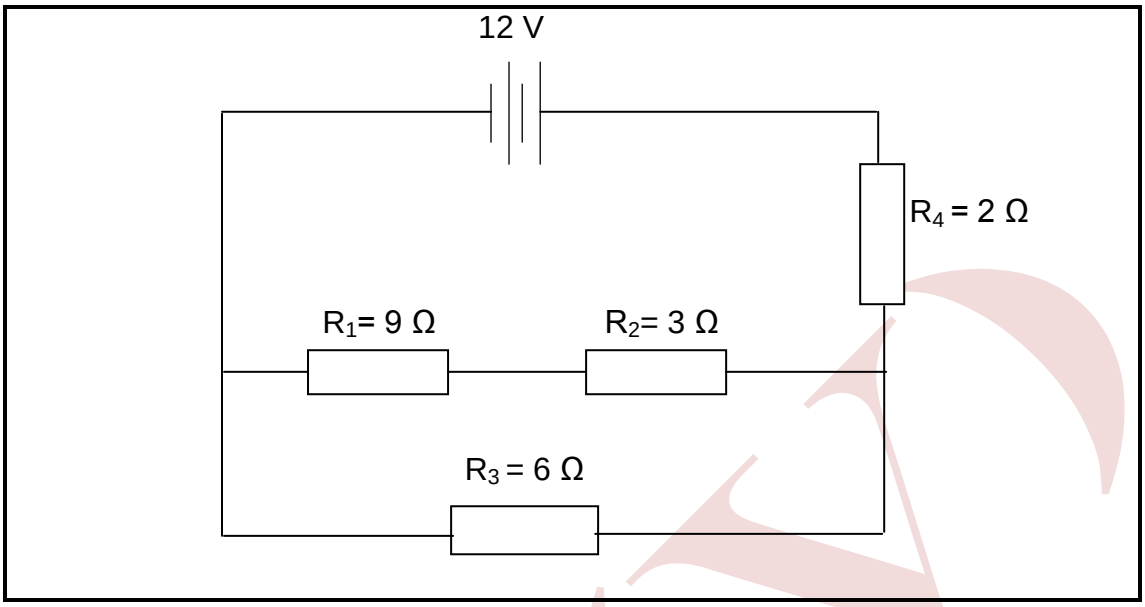


FIGURE 1

Refer to the circuit diagram in FIGURE 1 and calculate:

3.1.1 The total resistance of the circuit (4)

3.1.2 The total current of the circuit (2)

3.1.3 The current flowing through the 9 Ω resistor (4)

3.1.4 The total current of the circuit if the 3 Ω resistor is open-circuit (4)

3.2 Explain each of the following terms:

3.2.1 Emf of cells

3.2.2 Terminal voltage

(2 × 2) (4)

3.3 Three inductors of 20 mH, 4 mH and 5 mH are connected in parallel.

Calculate the total inductance. (3)

3.4 Two capacitors of 30 µF and 10 µF are connected in series across a 12 V supply.

Calculate:

3.4.1 The total capacitance (2)

3.4.2 The potential difference across the 10 µF capacitor (4)

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3.5 Capacitors are available in various sizes and types.

Name TWO types of polarised capacitors. (2)

3.6 Transformers play an important role in the electrical-distribution network.

Which type of transformer has more windings on the primary side than on the secondary side? (1)
[30]

QUESTION 4

4.1 Moving-coil and moving-iron instruments are used to measure the voltage and current of circuits.

Give TWO advantages of moving-coil instruments. (2)

4.2 Explain each of the following devices in analogue instruments:

4.2.1 Damping

4.2.2 Controlling

(2 × 2) (4)

4.3 There are two types of moving-iron instruments, namely attraction and repulsion types.

Describe the operation of a repulsion-type moving-iron instrument. (3)

4.4 Most analogue meters are sensitive and accurate. However, the accuracy of a good meter can be wasted by reading it incorrectly.

Explain the concept of *parallax error*. (2)

4.5 A moving-coil instrument gives a full-scale deflection if 15 mA flows through it. The resistance of the coil is 5 Ω.

Calculate the value of the multiplier resistor required to use a voltmeter up to 10 V. (3)

4.6 Explain how a frequency meter is connected in a circuit. (1)
[15]

QUESTION 5

5.1 DC motors have many practical applications in nearly all engineering and domestic fields.

List THREE practical applications of a series motor. (3)

5.2 Refer to FIGURE 2 and answer the questions.

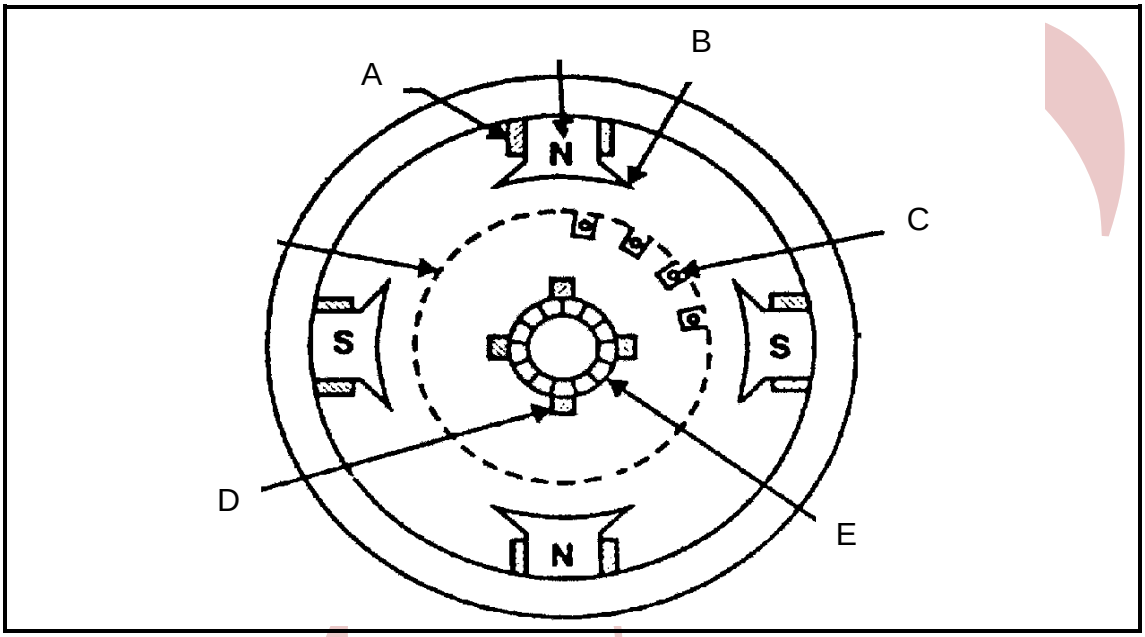


FIGURE 2

5.2.1 Identify FIGURE 2. (1)

5.2.2 Identify the parts by writing only the answer next to the letter (A–E) in the ANSWER BOOK. (5)

5.3 A universal motor operates on both AC and DC power supplies.

What are the FOUR main parts of a universal motor? (4)

5.4 A shunt-wound generator with a shunt field resistance of 150 ohms and an armature resistance of 0,05 ohms supplies 55 amperes at a terminal voltage of 250 volts to a certain load.

Calculate the generated emf. (5)

5.5 Capacitor-start/capacitor-run motors have excellent starting torque of about 300% of the full-load torque.

Give TWO applications of capacitor-start/capacitor-run motors. (2)
[20]

TOTAL: 100

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FORMULA SHEET

1. $Cos\phi = \frac{R}{Z}$

2. $P = V\ I\ Cos\phi$

3. $Q = V\ I\ Sin\phi$

4. $S = V\ I$

5. $V_1 = \left(\frac{R_1}{R_t}\right) V_T$

6. $I_1 = \left(\frac{R_2}{R_1 + R_2}\right) I_T$

7. $R_T = R_1 + R_2 + R_3 + \cdots + R_n$

8. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$

9. $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

10. $V_L = \sqrt{3}\, V\ p\ h$

11. $E = V - I_a R_a$

12. $f = \frac{Np}{60}$

13. $S = \frac{N_s - N_r}{N_s}$

14. $R_{sh} = \frac{I_m R_m}{I_{sh}}$

15. $R_{se} = \frac{V_t}{I_t} - R_m$

16. $F = B\ I\ l$

17. $\tan\phi = \sqrt{3}\left(\frac{W_1 - W_2}{W_1 + W_2}\right)$

18. $E = V + I\ r$

19. $V = I\ R$

20. $R = \frac{p\ l}{A}$

21. $R_t = R_0(1 + \alpha t)$

22. $R_t = R_0[1 + \propto_0 (t - \emptyset)]$

23. $P = V\ I$

24. $E = I^2\ R\ t$

25. $\eta = \frac{P_0}{P_{in}} \times 100$

26. $X_L = 2\pi\ f\ L$

27. $X_C = \frac{1}{2\pi\ f\ C}$

28. $Z = \sqrt{R^2 + (X_L - X_C)^2}$

29. $Mmf = N\ I$

30. $H = \frac{Mmf}{l}$

31. $B = \frac{\Phi}{A}$

32. $E = B\ l\ v$

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33. $E = -L\frac{di}{dt}$

34. $A = \frac{\pi d^2}{4}$

35. $E = \frac{1}{2} C V^2$

36. $Q = C V$

37. $F = 2\times10^{-7} \times \frac{I_1 I_2}{d}$

38. $I = \frac{Emf}{R + r_T}$

39. $L = N\frac{\Delta \phi}{\Delta I}$

40. $E = V + R_a I_a$

41. $E = -N\frac{d\emptyset}{dt}$

42. $E = \frac{V}{d}$

43. $E = \frac{1}{2} LI^2$

44. $Q = It$

45. $C_T = C_1 + C_2 + C_3 + \cdots + C_n$

46. $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$

47. $L_T = L_1 + L_2 + L_3 + \cdots + L_n$

48. $\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_n}$

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